Lesson 2 · How machines learn from examples

Teach the machine

Examples, labels, training and testing

Name _____
Class _____
Date _____

Work with a partner. Read each activity and answer.

1 Label the examples

You are labelling training data. Tick one box for each message.

	Safe message	Scam
a Football practice moved to 4 pm	☐	☐
b You won free airtime. Send your PIN now	☐	☐
c Your account will close. Click this link	☐	☐
d Exam timetable is on the notice board	☐	☐
e Scholarship approved. Pay a fee to claim it	☐	☐
f Library books are due back on Friday	☐	☐

2 Put it in order

Write 1 to 5 in the boxes to show how a machine learns.

☐ Test on new examples

☐ Collect examples

☐ Use: make predictions

☐ Label them

☐ Train: find patterns

3 Finish the sentence

Use words from the box.

Word box: **patterns** **model** **test** **rules**

The computer finds _____ in the training data and builds a _____, which we _____ on new examples.

Challenge

A spam filter trained on only 10 examples gets many new messages wrong. What two things could help it?

_____

_____